

Tyler Kuipers

SOFTWARE ARCHITECT

Edmonton, Canada

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About Me

I've spent 10 years building backend systems and infrastructure, with the last 3 focused on architecture and technical leadership. I design distributed, event-driven systems: APIs serving 5 teams with 99.9% uptime processing \$4.3M in transactions, microservices handling millions of asynchronous events, and a current rewrite of a legacy document generation platform into containerized C# microservices on Kubernetes. Earlier in my career I built backend services for a medical imaging AI SaaS at Arterys, used in clinical settings. My day-to-day is Python, Java, and TypeScript, with hands-on C# on the document generation work, plus Postgres and Kubernetes underneath. I lead through technical vision: RFCs, code reviews, mentorship, and a seat on AMA's Architecture Steering Committee.

Work Experience

Alberta Motor Association

SOFTWARE ARCHITECT

Edmonton, Alberta

Jan 2023 - Present

- Leading modernization of legacy document generation system into containerized C# microservices on EKS. 4-5 services (with sidecars) on a 3-5 node cluster, deployed via Helm. Datadog observability for failure rate, success count, and end-to-end latency. Leading 2-3 engineers from architecture through production. (C#, Kubernetes, EKS, Docker, Helm, Datadog)
- Built self-service developer platform solo that reduced environment provisioning from 2-3 weeks to 2 hours (60x improvement). Adopted by 15 engineers across 3 teams, created 120+ environments. Unblocked major initiatives by enabling rapid iteration and production debugging. (Python, Terraform, AWS, Jenkins, Docker)
- Own API architecture and SLOs for platform serving 5 internal teams and external partners. Designed API contracts, approve endpoint additions, maintain OpenAPI standards, and ensure consistency across services. APIs process \$4.3M in annual transactions. (Python, REST, OpenAPI, TypeScript)
- Architected event-driven cloud infrastructure for Canada's first pay-as-you-go insurance product across multiple cloud accounts. 10 serverless functions, \$4.3M annual revenue, 2.8M events processed, 99.9% uptime. Won 2024 Guidewire Innovation Award. (Python, AWS Lambda, Terraform, TypeScript)
- Set technical standards and provide architectural guidance for 12-person team through code reviews, RFC authorship, and technical documentation. Member of Architecture Steering Committee. Mentor engineers on system design, debugging, and best practices.

Arterys Inc. (Acquired by Tempus Labs)

SOFTWARE DEVELOPER

Calgary, Alberta (Remote)

Aug 2021 - Aug 2022

- Built and operated backend services for a medical imaging AI SaaS used in clinical settings, optimizing for low latency and high throughput on cloud infrastructure. (TypeScript, Node.js, AWS)
- Reduced AI algorithm processing turnaround using Redis caching and infrastructure tuning.
- Monitored production systems via Elasticsearch and Kibana, catching and resolving issues before customer impact.

Alberta Motor Association

SOFTWARE DEVELOPER | SOFTWARE TEAM LEAD

Edmonton, Alberta

Jun 2018 - Feb 2021 | Feb 2021 - Aug 2021

- Designed and built automated test framework from scratch that reduced defects by 30% on subsequent releases. Adopted across the insurance engineering team (15 developers) and still in use 4+ years later. Significantly improved developer velocity and release confidence. (Java/Gosu, JUnit)
- Led 5-6 person development team using TDD methodology. Delivered projects on time with no critical defects post-launch. Conducted code reviews, mentored junior developers, and collaborated with stakeholders on requirements and priorities.
- Led Git migration for 12-15 developers transitioning from SVN. Created Docker image for local dev environments, reducing onboarding friction for new developers.

Pleasant Solutions

SOFTWARE DEVELOPER

Edmonton, Alberta

Jan 2018 - Jun 2018

- Built HMI for a concrete plant control system using TDD methodology. Daily work in C# / .NET on an MVC architecture, with REST integration and unit testing in NUnit and Moq. (C#, .NET, MVC, REST)

Lockheed Martin Canada

SOFTWARE ENGINEER ASSOCIATE

Halifax, Nova Scotia

Jan 2017 - Jan 2018

- Contributed to Data Fusion module for Combat Management Systems, combining real-time data streams from multiple sensor sources. (Java, SQL)
- Designed and implemented scenario recorder system for capturing and replaying test scenarios, improving debugging and test coverage.

- Built web app for spatial/temporal data visualization (Edmonton River Atlas). (PHP, JavaScript, PostgreSQL, PostGIS)
- Wrote undergrad thesis on spatial-temporal referencing methods.
- Presented research through poster presentations.

Education

The Open University MSC. COMPUTING (SOFTWARE ENGINEERING) Currently pursuing Master's in Computing at The Open University.	Milton Keynes, England (Remote) May 2024 - Ongoing
The King's University B.S. IN COMPUTER SCIENCE	Edmonton, Alberta Sep 2012 - Dec 2016

Skills

Backend Architecture & Distributed Systems	Design event-driven, distributed backends with strong async workflows. Built systems handling millions of events through queues, serverless functions, and microservices on Kubernetes. Comfortable with distributed system trade-offs, scaling, and operational reliability. APIs serving 5 teams with 99.9% uptime. Own REST API contracts and SLOs for production APIs processing \$4.3M in annual transactions. Maintain OpenAPI standards across services, design versioning strategies, and approve endpoint additions for 5 internal teams and external partners. Long-running practitioner of TDD and code review discipline.
APIs & Service Design	Production AWS across multiple accounts (Lambda, EKS, SQS, RDS, IAM) with infrastructure as code in Terraform. Built developer platforms (60x faster provisioning), production EKS clusters, and observability with Datadog. Currently leading a C# / Kubernetes microservices rewrite of a legacy document generation system.
Cloud & Microservices Platforms	Own technical domains end-to-end: architecture, operations, and SLOs. Set standards for 12-person team through RFC authorship, code reviews, and mentorship. Member of Architecture Steering Committee. Lead projects from design through production with cross-functional collaboration.
Cross-Team Leadership	

Technologies

Backend Languages	Python, Java, TypeScript, C#, Node.js, Bash
Architecture & APIs	Event-Driven Architecture, Microservices, REST APIs, OpenAPI, Async Workflows
Databases	PostgreSQL, Aurora/RDS, Redis
Cloud & Infrastructure	AWS (Lambda, EKS, EC2, RDS, S3, SQS, IAM), Terraform, Kubernetes, Docker, Helm
Async & Messaging	SQS, Lambda, Event-Driven Patterns, Distributed Workflows
Observability	Datadog, Elasticsearch, Kibana, Metrics Design, Alerting